

LAZARD LEGACY MATH ACADEMY

College Algebra Essentials

Quick Drill — 30 Minutes

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"Dreams can come true. Better late than never."

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TABLE OF CONTENTS

1. Quick Review: Functions & Notation
2. Quick Review: Solving Equations & Inequalities
3. Quick Review: Factoring & Polynomials
4. Practice Drill: 30 College Algebra Problems
5. Answer Key
6. Student Notes

Section 1: Quick Review — Functions & Notation

What is a Function?

A function is a rule that assigns exactly ONE output to each input. We write it as $f(x)$ — read as 'f of x.'

Key Vocabulary:

- **$f(x)$** = function notation. The x is the input, $f(x)$ is the output.
- **Domain** = all valid input values (x -values that work).
- **Range** = all possible output values (y -values produced).
- **Composition**: $f(g(x))$ means 'plug $g(x)$ into f .'

Domain Restrictions to Watch For:

- Cannot divide by zero $\rightarrow$ denominator $\neq 0$
- Cannot take square root of negative $\rightarrow$ expression under $\sqrt{}$ must be ≥ 0

Worked Examples:

Example 1:

If $f(x) = 3x - 5$, find $f(4)$.

Solution: Replace every x with 4: $f(4) = 3(4) - 5 = 12 - 5 = 7$.

Example 2:

If $g(x) = x^2 + 2x - 1$, find $g(-3)$.

Solution: $g(-3) = (-3)^2 + 2(-3) - 1 = 9 + (-6) - 1 = 2$.

Example 3:

If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(2))$.

Solution: Step 1: Find $g(2) = 2^2 = 4$. Step 2: Find $f(4) = 2(4) + 1 = 9$. Answer: $f(g(2)) = 9$.

Example 4:

Find the domain of $f(x) = 1/(x - 3)$.

Solution: The denominator cannot be zero: $x - 3 \neq 0$, so $x \neq 3$. Domain: **all real numbers except $x = 3$** .

Example 5:

If $h(x) = \sqrt{x + 4}$, find the domain.

Solution: Expression under $\sqrt{}$ must be ≥ 0 : $x + 4 \geq 0 \rightarrow x \geq -4$. Domain: **$[-4, \infty)$** .

Section 2: Quick Review — Solving Equations & Inequalities

Methods for Solving Quadratics ($ax^2 + bx + c = 0$):

1. **Factoring:** Write as $(x + m)(x + n) = 0$, then set each factor = 0.
2. **Square Root Method:** If $x^2 = k$, then $x = \pm\sqrt{k}$.
3. **Quadratic Formula:** $x = [-b \pm \sqrt{(b^2-4ac)}] / (2a)$.

Inequality Rules:

- Solve like an equation, BUT flip the inequality sign when multiplying or dividing by a **NEGATIVE** number.
- Absolute value inequalities: $|x| \leq a$ means $-a \leq x \leq a$.

Worked Examples:

Example 1:

Solve: $2x^2 - 8 = 0$

Solution: Add 8: $2x^2 = 8$. Divide by 2: $x^2 = 4$. Take square root: $x = \pm\sqrt{4} = \pm 2$.

Example 2:

Solve: $x^2 + 3x - 10 = 0$

Solution: Factor: $(x + 5)(x - 2) = 0$. Set each = 0: $x = -5$ or $x = 2$.

Example 3:

Solve using the quadratic formula: $x^2 - 4x + 1 = 0$

Solution: $a=1, b=-4, c=1$. $x = [4 \pm \sqrt{(16-4)}] / 2 = [4 \pm \sqrt{12}] / 2 = 2 \pm \sqrt{3} \approx 3.73$ or 0.27 .

Example 4:

Solve the inequality: $3x + 2 > 11$

Solution: Subtract 2: $3x > 9$. Divide by 3: $x > 3$. Solution: **$(3, \infty)$** .

Example 5:

Solve: $|2x - 5| \leq 3$

Solution: Remove absolute value: $-3 \leq 2x - 5 \leq 3$. Add 5: $2 \leq 2x \leq 8$. Divide by 2:
 $1 \leq x \leq 4$ or $[1, 4]$.

Section 3: Quick Review — Factoring & Polynomials

Factoring Methods (in order of what to try first):

1. **GCF (Greatest Common Factor):** Always check this FIRST.
2. **Difference of Squares:** $a^2 - b^2 = (a + b)(a - b)$.
3. **Trinomial ($x^2 + bx + c$):** Find m, n where $m \times n = c$ and $m + n = b$.
4. **AC Method ($ax^2 + bx + c$ when $a \neq 1$):** Find factors of $a \times c$ that add to b .
5. **Sum/Difference of Cubes:** $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$.

Worked Examples:

Example 1:

Factor: $4x^3 - 12x^2$

Solution: Find GCF: $4x^2$. Factor out: $4x^2(x - 3)$.

Example 2:

Factor: $x^2 - 25$

Solution: Difference of squares: $(x + 5)(x - 5)$.

Example 3:

Factor: $x^2 + 7x + 12$

Solution: Find m, n where $m \times n = 12$ and $m + n = 7$. Answer: 3 and 4. Result: $(x + 3)(x + 4)$.

Example 4:

Factor: $2x^2 + 5x - 3$

Solution: AC method: $a \times c = -6$. Factors of -6 that add to 5: 6 and -1. Group: **$(2x - 1)(x + 3)$** .

Example 5:

Expand: $(x + 2)(x^2 - 2x + 4)$

Solution: Multiply each term: $x^3 - 2x^2 + 4x + 2x^2 - 4x + 8 = \mathbf{x^3 + 8}$. (Sum of cubes pattern.)

Section 4: Practice Drill — 30 College Algebra Problems

Directions: Solve each problem. Show your work in the space provided.

Functions (1–8):

1. If $f(x) = 5x - 2$, find $f(3)$.
2. If $g(x) = x^2 - 4x + 1$, find $g(5)$.
3. If $f(x) = 2x + 3$, find $f(-1)$.
4. Find the domain of $f(x) = \sqrt{2x - 6}$.
5. Find the domain of $g(x) = 1/(x^2 - 4)$.
6. If $f(x) = x^2 + 1$, find $f(a + 1)$.
7. If $f(x) = 2x + 1$ and $g(x) = x - 3$, find $f(g(x))$.
8. If $h(x) = 3x - 7$, find $h(0)$ and $h(5)$.

Equations & Inequalities (9–18):

9. Solve: $x^2 - 16 = 0$.

10. Solve: $x^2 + 6x + 8 = 0$.

11. Solve: $2x^2 - 5x - 3 = 0$.

12. Solve: $x^2 - 2x - 15 = 0$.

13. Solve using quadratic formula: $x^2 + 2x - 5 = 0$.

14. Solve: $4x - 1 > 15$.

15. Solve: $|x + 3| = 8$.

16. Solve: $|3x - 2| \leq 7$.

17. Solve: $2x + 5 \leq 3x - 1$.

18. Solve: $(x - 2)(x + 5)(x - 1) = 0$.

Factoring & Polynomials (19–30):

19. Factor: $6x^2 - 18x$.

20. Factor: $x^2 - 49$.

21. Factor: $x^2 + 9x + 20$.

22. Factor: $x^2 - 5x - 14$.

23. Factor: $3x^2 + 11x + 6$.

24. Factor: $4x^2 - 1$.

25. Simplify: $(2x^2 + 3x - 5) + (x^2 - 4x + 2)$.

26. Simplify: $(3x^2 + 2x) - (x^2 - 5x + 3)$.

27. Multiply: $(x + 3)(x - 7)$.

28. Multiply: $(2x - 1)(x + 4)$.

29. Solve: $x^3 - 4x = 0$.

30. Factor completely: $x^3 - 8$.

Section 5: Answer Key

1. $f(3) = 5(3) - 2 = 13$
2. $g(5) = 25 - 20 + 1 = 6$
3. $f(-1) = 2(-1) + 3 = 1$
4. $2x - 6 \geq 0 \rightarrow x \geq 3$. Domain: $[3, \infty)$
5. $x^2 - 4 \neq 0 \rightarrow x \neq \pm 2$
6. $f(a+1) = (a+1)^2 + 1 = a^2 + 2a + 2$
7. $f(g(x)) = 2(x-3) + 1 = 2x - 5$
8. $h(0) = -7$, $h(5) = 8$
9. $x^2 = 16 \rightarrow x = \pm 4$
10. $(x+2)(x+4) = 0 \rightarrow x = -2$ or -4
11. $(2x+1)(x-3) = 0 \rightarrow x = -1/2$ or 3
12. $(x-5)(x+3) = 0 \rightarrow x = 5$ or -3
13. $x = [-2 \pm \sqrt{24}]/2 = -1 \pm \sqrt{6}$
14. $4x > 16 \rightarrow x > 4$
15. $x = 5$ or $x = -11$
16. $-7 \leq 3x-2 \leq 7 \rightarrow -5/3 \leq x \leq 3$
17. $x \geq 6$
18. $x = 2$, -5 , or 1
19. $6x(x - 3)$
20. $(x+7)(x-7)$
21. $(x+4)(x+5)$
22. $(x-7)(x+2)$
23. $(3x+2)(x+3)$
24. $(2x+1)(2x-1)$
25. $3x^2 - x - 3$

26. $2x^2 + 7x - 3$

27. $x^2 - 4x - 21$

28. $2x^2 + 7x - 4$

29. $x(x+2)(x-2) \rightarrow x = 0, 2, -2$

30. $(x-2)(x^2+2x+4)$

